PROBLEM SET 2, PART 1: THE DIFFERENTIAL
DUE: OCT. 17

(1) [Not required] [**Tangent vectors defined by curves**]
There is a more geometric way to define tangent vectors: Take a chart $\{\varphi, U, V\}$ around p. Let $\mathcal{C}_p$ be the set of all smooth maps $\gamma : (-\varepsilon, \varepsilon) \to U$ such that $\gamma(0) = p$. Define an equivalence relation on $\mathcal{C}_p$ by

$$\gamma_\alpha \sim \gamma_\beta \iff \frac{d(\varphi \circ \gamma_\alpha)}{dt}(0) = \frac{d(\varphi \circ \gamma_\beta)}{dt}(0).$$

Define a natural vector space structure on $\mathcal{C}_p/\sim$ so that T_pM is isomorphic to $\mathcal{C}_p/\sim$.

(2) [**Product of manifolds – continued**]
Let M, N be smooth manifolds and consider the product $M \times N$. Prove:
(a) The nature projections $\pi_1 : M \times N \to M$ and $\pi_2 : M \times N \to N$ are smooth.
(b) [Universality] If P is any smooth manifold, then a map $f : P \to M \times N$ is smooth if and only if both $\pi_1 \circ f$ and $\pi_2 \circ f$ are smooth.
(c) $T_{(p_1,p_2)}(M_1 \times M_2) \simeq T_{p_1}M_1 \oplus T_{p_2}M_2$.
(d) More generally, let $f_1 : M_1 \to N_1$ and $f_2 : M_2 \to N_2$ be smooth maps, and $p_1 \in M_1, p_2 \in M_2$. Find the relation between $d(f_1 \times f_2)_{(p_1,p_2)}$ and $(df_i)_{p_i}$, $i = 1, 2$.

(3) [**The tangent bundle**]
Let M be a smooth manifold of dimension n. Let $TM = \bigcup_p T_pM$ be the disjoint union of all tangent vectors. We will call TM the *tangent bundle* of M. We will denote an element X_p in TM by (p, X_p), to emphasis its "base point". There is a natural projection map

$$\pi : TM \to M, \quad (p, X_p) \mapsto p.$$

For each chart (φ, U, V) of M, we define a bijective map

$$T\varphi = (\varphi \circ \pi, d\varphi) : \pi^{-1}(U) \to V \times \mathbb{R}^n, \quad (p, X_p) \mapsto (\varphi(p), d\varphi_p(X_p)).$$

We endow a topology on TM so that each $T\varphi$ is a homeomorphism. Prove:
(a) $(T\varphi, \pi^{-1}(U), V \times \mathbb{R}^n)$ is a chart of TM.
(b) Charts of this type are compatible, so TM is a $2n$-dimensional smooth manifold.
(c) No mater M is orientable or not, the tangent bundle TM is orientable.
(d) The differential $d\pi_{(p,X_p)} : T_{(p,X_p)}TM \to T_pM$ is always surjective.
(e) TS^1 is diffeomorphic to $S^1 \times \mathbb{R}$.

(4) [Not required] [**Submersions are open maps**]
(a) Let $f : M \to N$ be a submersion. Prove: f is an open map.

(b) Prove: If M is compact and N is connected, then any submersion $f : M \to N$ is surjective. Conclude that there exists no submersion from any compact smooth manifold M to any connected noncompact smooth manifold (e.g. $\mathbb{R}^n$).

(5) [**Hadamard's global inverse function theorem**]
Let M, N be connected smooth manifolds, and $f \in C^\infty(M, N)$.
(a) Prove: If f is proper (i.e. the pre-images of compact subsets are compact), then it is closed (i.e. the image of closed subsets are closed).
(b) [Not required] Prove: If f is proper and local diffeomorphism everywhere, then it is a covering map.
(c) Finally conclude that if f is proper and local diffeomorphism everywhere, and N is simply connected, then f is a diffeomorphism.

(6) [**Compositions of submersion/immersions/constant rank maps**]
Consider the compositions of submersions/immersions/constant rank maps. [We assume all maps are smooth.]
(a) Is it true that the composition of two submersions is still submersion? What about the composition of immersions? What about constant rank maps?
(b) What if we composite a constant rank map with a submersion? With an immersion? A submersion with an immersion?
(c) If the composition $g \circ f$ is an immersion, can we conclude f is an immersion?
(d) If the composition $g \circ f$ is a submersion, can we conclude g is a submersion? If not, what extra condition do we need?

(7) [**Matrix Lie groups**]
Recall that $\mathrm{GL}(n, \mathbb{R})$ is an n^2 dimensional smooth manifold.
(a) Consider the map $\det : \mathrm{GL}(n, \mathbb{R}) \to \mathbb{R}$.
  (i) Prove: det is a smooth function.
  (ii) For any $X \in \mathrm{GL}(n, \mathbb{R})$, what is $T_X(\mathrm{GL}(n, \mathbb{R}))$?
  (iii) Show that $(d \det)_X(A) = (\det X)\mathrm{tr}(X^{-1}A)$.
  (iv) Show that det is a submersion.
(b) [Not required] Consider the map

$$f : \mathrm{GL}(n, \mathbb{R}) \to \mathrm{GL}(n, \mathbb{R}), \quad X \mapsto f(X) = X^T X.$$

  (i) Prove: f is smooth, and $df_X(A) = X^T A + A^T X$.
  (ii) Prove: f has constant rank $n(n+1)/2$.

(8) [Not required] [**Germ and the category of pointed manifolds**]
Suppose you are a teacher. Write down lecture notes (no more than 1 page), starting with the definition of germ below (and explore some properties of germ that you may need), then explain what is "the category of pointed smooth manifolds" and why "taking differential at a point" is a functor.

Definition of germ: Let M, N be smooth manifolds and $p \in M$. Let $f_i : U_i \to N$ be smooth maps, where U_i are neighborhoods p. For such

functions, define an equivalence relation by

$$f_1 \sim f_2 \iff \exists U \subset U_1 \cap U_2 s.t. f_1|_U = f_2|_U.$$

The equivalence class containing f is called an *germ* of smooth maps at p and is denoted by $[f]_p$.

Here are some necessary definitions of category and functor you may assume without repeating:

定义 1.148.（范畴）

一个**范畴** $\mathcal{C}$ 包含

1. 一个类 $\mathrm{Ob}(\mathcal{C})$，其中的元素称为 **对象**,
2. 一个类 $\mathrm{Mor}(\mathcal{C})$，其中的元素称为对象间的 **态射**，满足
   - 每个态射 f 都有一个 **始对象** $X \in \mathrm{Ob}(\mathcal{C})$ 和一个 **终对象** $Y \in \mathrm{Ob}(\mathcal{C})$.
     - 我们记 $f: X \to Y$ 并称 “f 是从 X 到 Y 的态射”.
     - 我们将从 X 到 Y 的态射全体记为 $\mathrm{Mor}(X, Y)$. [a]
   - 态射 $f: X \to Y$ 和 $g: Y \to Z$ 的复合是态射 $g \circ f: X \to Z$，且满足

     (a). (结合性) 设 $f: X \to Y, g: Y \to Z$ 和 $h: Z \to W$, 则

     $$h \circ (g \circ f) = (h \circ g) \circ f,$$

     (b). (单位元) 对 $X \in \mathrm{Ob}(\mathcal{C})$, 存在 **单位态射** $\mathrm{Id}_X : X \to X$, 使得对于任意态射 $f: Z \to X$ 和 $g: X \to Y$，都有

     $$\mathrm{Id}_X \circ f = f \quad \text{且} \quad g \circ \mathrm{Id}_X = g.$$

[a] 对于一些对象 X 和 Y, 可能并不存在从 X 到 Y 的态射, 此时 $\mathrm{Mor}(X, Y) = \emptyset$.

定义 3.2.20.（函子）

设 $\mathcal{C}, \mathcal{D}$ 是范畴. 从 $\mathcal{C}$ 到 $\mathcal{D}$ 的 (协变) **函子**[a] F 是一个对应，使得

(1) 将 $\mathrm{Ob}(\mathcal{C})$ 中的每个对象 X 都对应到 $\mathrm{Ob}(\mathcal{D})$ 中的某个对象 $F(X)$,

(2) 将范畴 $\mathcal{C}$ 中的每个态射 $f \in \mathrm{Mor}(X, Y)$ 都对应到范畴 $\mathcal{D}$ 中的某个态射 $F(f) \in \mathrm{Mor}(F(X), F(Y))$, 且该对应满足：

- 对范畴 $\mathcal{C}$ 中的任意对象 $X \in \mathrm{Ob}(\mathcal{C})$，有

$$F(\mathrm{Id}_X) = \mathrm{Id}_{F(X)},$$

- 对范畴 $\mathcal{C}$ 中的任意态射 $f \in \mathrm{Mor}(X, Y)$ 和 $g \in \mathrm{Mor}(Y, Z)$ 都有

$$F(g \circ f) = F(g) \circ F(f).$$

[a] 类似地我们可以定义**反变函子**的概念：反变函子也是一个对应，它将每个态射 $f \in \mathrm{Mor}(X, Y)$ 对应到态射 $F(f) \in \mathrm{Mor}(F(Y), F(X))$ 并且满足 $F(\mathrm{Id}_X) = \mathrm{Id}_{F(X)}$ 和 $F(g \circ f) = F(f) \circ F(g)$.